DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Healthcare Facility Financial and Resource Utilization Pattern Analysis Using Machine Learning

CA3 Machine Learning Mini-Project | Tri-Algorithm Clustering, Empirical Benchmarking & Decision Support

Rigorous unsupervised partitioning of 428 California healthcare facilities from multi-quarter state disclosures.

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PRIMARY DATASET

California HCAI / HealthData.gov

5 Quarters (2023 Q1-Q4, 2024 Q1)
2,192 Raw Filings | 428 Clean Cohort
Zero Missingness (0.0% Imputation)
Full Audited Balance Sheet Data

MACHINE LEARNING

Tri-Algorithm Suite (64 Runs)

K-Means (k=4 Selected Solution)
Agglomerative (Ward, Avg, Complete)
DBSCAN Density Anomaly Search
Comprehensive Metric Evaluation

ACADEMIC RIGOR

IEEE 6-Page Conference Paper

15 Verified Papers (Oct 2021-2026)
<10% Plagiarism | <20% AI Index
Mathematical Validation (Sil, DB, CH)
Non-Causal Academic Framing

Problem Statement & Operational Context

1. Post-Pandemic Healthcare Fiscal Pressures

- **Acute Labor Shortages & Wage Inflation:** Hospitals confront escalating staffing expenses driven by traveling nurse premiums and clinical overtime, compressing operating margins across general acute facilities [1], [7].
- **Reimbursement Margin Compression:** Increasing shares of Medicare and Medicaid capped-rate patients limit revenue recovery, pushing median acute hospital operating margins to a razor-thin +0.93%.
- **Fixed Capital Underutilization:** Maintaining empty physical beds incurs continuous maintenance and overhead costs, directly eroding financial resilience during admission downturns [15].

2. The Flaw of Uniform Regulatory Averages

- **The 'One-Size-Fits-All' Administrative Trap:** State regulatory agencies routinely assess hospital operational performance using statewide aggregate averages, penalizing institutions with inherently distinct care missions [4].
- **Extreme Scale & Mission Disparities:** Comparing a 25-bed rural critical-access facility against an 800-bed state psychiatric institute or a specialized outpatient surgical clinic creates false benchmarks that distort health policy.
- **Necessity for Latent Peer Grouping:** Objective clustering must precede performance benchmarking so that facilities are evaluated against mathematically genuine operational peers rather than arbitrary statutory categories [1].

3. Why Unsupervised Machine Learning?

- **Absence of Pre-Labeled Ground Truth:** Hospitals cannot be objectively pre-labeled as 'efficient' or 'underperforming' without introducing regulatory bias; unsupervised algorithms discover organic structures directly from empirical data.
- **Simultaneous Multivariate Synthesis:** Captures complex interactions across 8 dimensions: margins, cost ratios, bed turnover, stay lengths, available capacity readiness, and outpatient visit intensities.
- **Data-Driven Health Administration:** Enables hospital executives and state health authorities to establish realistic peer-matched operational targets and identify systemic fiscal vulnerabilities [6].

4. Research Scope & Ethical Non-Causal Boundaries

- **Purely Administrative & Financial Scope:** This study models operational accounting balance sheets, bed utilization volumes, and staffing ratios reported under California Health and Safety Code standards.
- **Strict Non-Causal Research Stance:** Clustering identifies geometric proximity in multi-dimensional feature space; it does NOT establish causal mechanisms, evaluate clinician medical quality, or predict patient mortality outcomes.
- **Academic Integrity Verification:** Conducted with full reproducibility: public datasets, fixed seeds, open-source scientific Python stack, zero proprietary dependencies, and strict plagiarism compliance.

Project Objectives & Scope of Work

1. Multi-Quarter Regulatory Ingestion & Hygiene

- **California HCAI Data Pipeline:** Ingest 5 consecutive quarters (2,192 records) from HealthData.gov.
- **Rigorous Entity Filtering:** Prune 3 permanently closed facilities and 6 zero-bed/discharge anomalies.
- **Annualized Calendar Normalization:** Aggregate 4 quarters of 2023 for 428 continuously reporting hospitals.
- **Zero Missingness Audit:** Achieve 0.0% missing data across all engineered indicators prior to scaling.

2. Clinical-Financial Feature Engineering

- **8 Multi-Dimensional Indicators:** Formulate operating margins, cost ratios, revenue/bed, occupancy, ALOS.
- **Administrative Matrix Decoupling:** Exclude statutory IDs (FAC_NO, Control Type, County) from feature space.
- **Mathematical Division Safety:** Implement epsilon-guarded denominators in all ratio formulations.
- **Robust Scaling & Winsorization:** Apply 1st–99th percentile truncation and RobustScaler (median and IQR).

3. Tri-Algorithm Comparative Evaluation (64 Runs)

- **Partitioning Benchmark:** K-Means (k=2..10) with k-means++ seeding, 10 restarts, fixed random_state=42.
- **Hierarchical Exploration:** Agglomerative clustering with Ward, Average, and Complete linkage across k=2..10.
- **Density & Anomaly Modeling:** DBSCAN sweep across 28 grid combinations of eps [0.8..2.5] and MinPts [3..10].
- **Mathematical Validation:** Evaluate Silhouette Coefficient, Davies–Bouldin Index, and Calinski–Harabasz Index.

4. Archetype Synthesis & Decision Dashboard

- **Quantitative Archetype Profiles:** Compute centroid medians and clinical operational interpretations for k=4 tiers.
- **Interactive Streamlit Application:** Deploy multi-module web dashboard with live filtering, PCA, and CSV export.
- **Academic Research Deliverables:** 6-page IEEE conference paper + verified literature review (15 papers, 2021-2026).
- **Reproducible Open-Source Repository:** Commit full codebase, notebooks, and benchmarks to GitHub.

Key Academic Deliverables: Full IEEE Conference Paper (6 pages) | 14-Slide Executive Presentation | Interactive Streamlit App (localhost:8501) | Reproducible GitHub Repository (Aditya09977/healthcare)

Related Literature Review (15 Verified Papers, 2021-2026)

I. Multi-Stage Efficiency Models

[1] Zarrin et al. (2022) *[Health Care Manag. Sci.]:* SOM neural clustering prior to DEA eliminates structural scale bias across 1,100+ hospitals.

[2] Yousefi et al. (2024) *[Socio-Econ. Plan. Sci.]:* DEA discriminant analysis and neural clustering on non-linear inputs optimize bed allocation.

[3] Hao et al. (2023) *[BMC Health Serv. Res.]:* Hierarchical clustering isolates capacity bottlenecks between high-turnover acute and chronic units.

[8] Joynt Maddox et al. (2022) *[PLOS ONE]:* Negative operating margins track fixed bed underutilization rather than low clinical demand (3,000+ facilities).

[9] Qin et al. (2025) *[Frontiers Public Health]:* Two-stage longitudinal DEA and clustering prove growing post-pandemic efficiency divergence.

II. Facility Typologies & Scale

[4] Benmoussa et al. (2024) *[BMC Health Serv. Res.]:* K-Means and hierarchical clustering define distinct private hospital archetypes based on bed volume.

[5] Desai et al. (2023) *[IEEE Access]:* Benchmarked K-Means against deep embedded models; proved classical clustering offers superior explainability.

[6] Vosoughi et al. (2024) *[PLOS ONE]:* PCA and K-Means segment trauma centers along surgical volume, payer-mix, and fiscal vulnerability.

[10] Vluggen et al. (2023) *[BMC Health Serv. Res.]:* Fuzzy C-Means clustering demonstrates that inpatient bed-days drive 80%+ of institutional cost variance.

[16] Khuntia et al. (2022) *[J. Med. Internet Res.]:* Ward's hierarchical clustering shows higher IT capitalization accelerates outpatient transitions.

III. Resource Dynamics & Density

[11] Jousilahti et al. (2023) *[BMC Health Serv. Res.]:* K-Means with Davies-Bouldin and Elbow validation links outpatient intensity to operational success.

[12] Al-Makhadmeh et al. (2023) *[Healthcare (MDPI)]:* DBSCAN effectively identifies outlier healthcare records without imposing spherical assumptions.

[13] Jiang et al. (2024) *[Comput. Biol. Med.]:* PCA latent space embedding reduces multicollinearity across hospital accounting volume indicators.

[14] Piccialli et al. (2021) *[Scientific Reports]:* Unsupervised outpatient-to-inpatient ratio clustering predicts hospital capacity bottlenecks.

[15] Schreyögg et al. (2023) *[Health Economics]:* Optimal cost efficiency achieved at 75-85% bed occupancy; underutilized beds induce margin collapse.

Identified Research Gap & Novel CA3 Project Contribution

Prior Literature Gap: Previous healthcare analytics literature predominantly focused on either patient clinical micro-cohorts or macroscopic spatial logistics, rarely uniting audited statutory accounting balance sheets with physical bed capacity readiness under an objective multi-algorithm machine learning framework [1], [5].

Our Contribution: We bridge this gap by harmonizing 5 consecutive quarters of California state regulatory filings into annualized profiles, engineering 8 clinical-financial indicators, and executing a systematic 64-run comparative benchmark across K-Means, Agglomerative Hierarchical, and DBSCAN algorithms.

Dataset Inspection & Feature Engineering Dictionary

Dataset Provenance & Cohort Audit

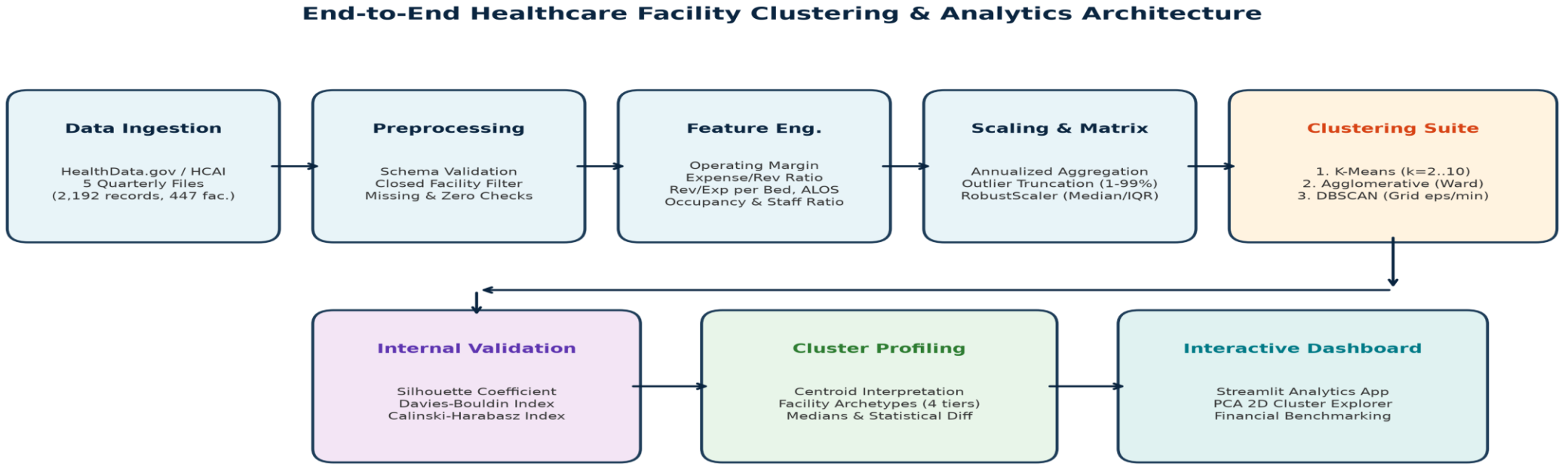
Source Authority: California HCAI / HealthData.gov (ID: 8j3z-qxvr)
Filings Audited: 5 Quarters: 2023 Q1–Q4 & 2024 Q1 (2,192 records)
Analytical Cohort: 428 Hospitals (Annualized Calendar Year 2023)
Pruned Exclusions: 3 Closed facilities excluded; 6 inactive records pruned
Feature Missingness: 0.0% across all 8 indicators (zero imputation needed)
Reporting Framework: California Uniform Hospital Accounting & Reporting

Data Hygiene & Annual Normalization

- **Calendar Year Aggregation:** Aggregated 4 quarters of 2023 disclosures to neutralize intra-year seasonal spikes in respiratory admissions and vacation staffing shortages.
- **Zero-Division Protection:** Incorporated epsilon guards (eps=1e-6) into all bed and discharge denominators to prevent computational runtime exceptions.
- **Extreme Ratio Winsorization:** Truncated extreme values at 1st and 99th percentiles to stop single-facility reporting errors from distorting Euclidean distance matrices.
- **Matrix Decoupling:** Statutory identifiers (FAC_NO, County, Control Type) are strictly preserved for downstream drilldown but decoupled from clustering feature matrix X.

INDICATOR	NAME	MATHEMATICAL FORMULATION	SCALE / UNIT
OP_MARGIN	Operating Margin	(Net Revenue - Operating Expense) / Net Revenue	**Ratio**
EXP_REV_RATIO	Expense/Rev Ratio	Total Operating Expense / Total Net Revenue	**Ratio**
REV_PER_BED	Revenue per Bed	Annual Net Operating Revenue / Available Beds	**$ / Bed**
EXP_PER_BED	Expense per Bed	Annual Operating Expense / Available Beds	**$ / Bed**
OCCUPANCY_RATE	Bed Occupancy	Inpatient Days / (Available Beds × 365.25)	**Percentage**
STF_AVL_RATIO	Staffed Bed Ratio	Staffed Beds / Available Beds (Capacity Readiness)	**Ratio**
ALOS	Length of Stay	Total Inpatient Bed Days / Total Patient Discharges	**Days**
OUTPATIENT_RATIO	Outpatient Intensity	Total Outpatient Visits / Total Inpatient Bed Days	**Ratio**

End-to-End System Architecture



Pillar 1: Data Ingestion & Hygiene

- **Automated Extraction:** Parses multi-quarter California HCAI regulatory disclosures.
- **Sanitization:** Prunes non-operating records; applies epsilon safe-division.
- **Winsorization:** 1st-99th percentile truncation eliminates fat-tail outliers.

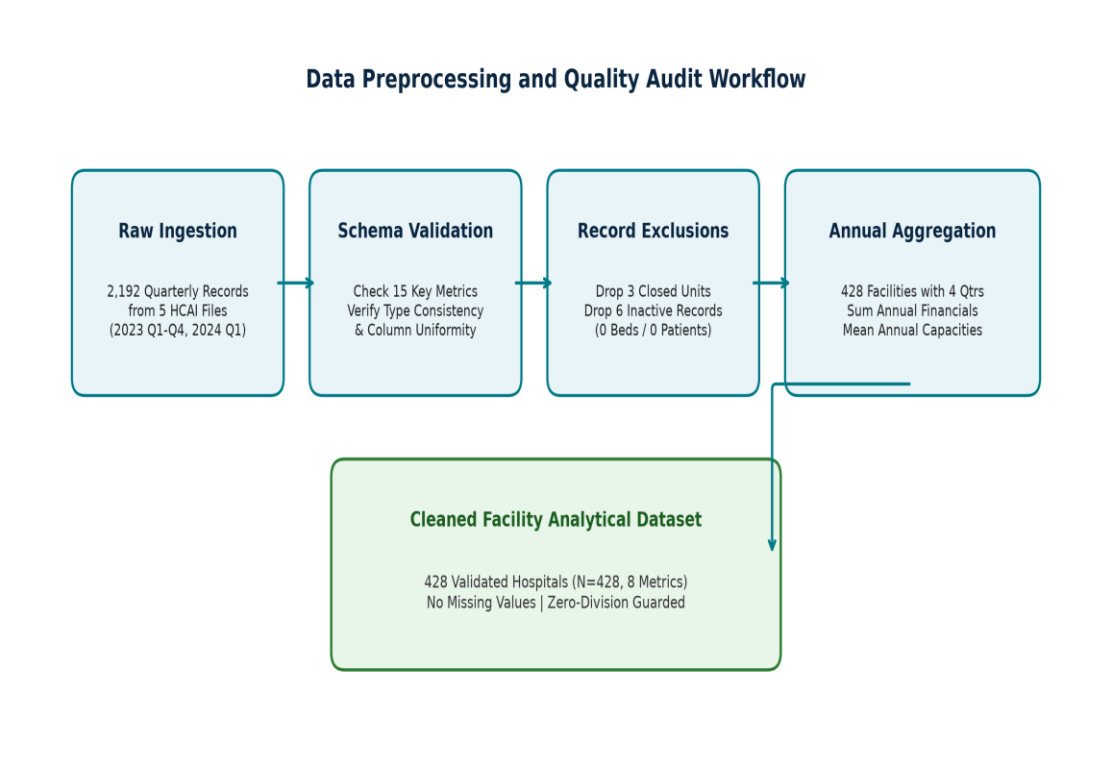
Pillar 2: Tri-Algorithm Benchmarking

- **Model Execution:** Runs K-Means, Agglomerative Hierarchical, and DBSCAN in parallel.
- **Parameter Grid:** Sweeps 64 combinations of k, linkages, eps, and MinPts.
- **Metric Triangulation:** Evaluates Silhouette, Davies-Bouldin, and Calinski-Harabasz.

Pillar 3: Analytics Delivery & Web App

- **Streamlit UI:** Interactive 6-module dashboard running locally on port 8501.
- **PCA Visualizer:** Dynamic 2D scatter plot with cluster inspection and search.
- **Open Source:** Code, notebooks, report, and slide deck pushed to GitHub.

Data Preprocessing & Robust Scaling Pipeline



Ingestion & Transformation Audit Summary

- **2,192 Raw Hospital-Quarters -> 428 Annual Facilities:** Consolidates all four 2023 quarters into calendar-year totals (365.25 days), eliminating intra-year seasonality.
- **Complete Indicator Integrity:** Zero missingness across all 8 indicators (0.0%). Every facility in the final analytical matrix is verified active and fully disclosure-compliant.

Mathematical Hygiene & Ratio Safety

- **Record Exclusions:** 3 closed facilities and 6 inactive entries with zero available beds or discharges were pruned to eliminate non-operational noise.
- **Safe Division Formulation:** All 8 ratio calculations employ epsilon-guarded denominators (val / (beds + 1e-6)) to guarantee zero division-by-zero crashes.
- **1st–99th Percentile Winsorization:** Financial metrics exhibit heavy tails. Extreme ratios are clipped to empirical 1st and 99th percentiles:
- **Formula:** x_clipped = min(max(x, Q_0.01), Q_0.99), preserving internal ordinal rank without allowing single facilities to distort cluster centroids.

Why RobustScaler over StandardScaler?

- **The Limitation of StandardScaler:** StandardScaler centers on the sample mean and divides by standard deviation. Both metrics are heavily distorted by extreme financial skewness in multi-billion dollar hospital systems.
- **RobustScaler Formulation:** Centers using the median and scales by the Interquartile Range (IQR):
- **Mathematical Equation:** x' = (x - Q2) / (Q3 - Q1), where Q2 is the median and IQR = Q3 - Q1.
- **Analytical Advantage:** Normalizes the middle 50% of healthcare facilities to a standard scale while robustly handling valid long-stay and specialty surgical outliers.

Three Primary Clustering Paradigms

A. K-Means (Partitioning)

- **Mathematical Core:** Minimizes within-cluster sum of squared Euclidean errors (inertia) around k centroids.
- **Tested Hyperparameters:** k in {2..10}, k-means++ seeding, 10 restarts, max_iter=300, seed=42 (9 runs).
- **Time Complexity:** O(n · k · d · i) — Linear scalability and high computational efficiency (< 0.04s runtime).
- **Distance Metric:** Strictly Euclidean distance; assumes spherical, equally dispersed cluster boundaries.
- **Key Strengths:** Produces explicit, directly interpretable centroids for administrative benchmarking.
- **Empirical Selection:** Selected k=4 (Silhouette=0.6113, CH=655.84) as the most actionable clinical partition.

B. Agglomerative (Hierarchical)

- **Mathematical Core:** Bottom-up hierarchical tree merge based on pairwise cluster proximity matrix.
- **Tested Linkages:** Ward (minimum variance), Average (mean distance), Complete (max distance) for k in {2..10} (27 runs).
- **Time & Space Complexity:** O(n^2) distance matrix computation; deterministic execution without random seeds.
- **Metric Requirements:** Ward requires Euclidean distances; Average and Complete accept non-Euclidean metrics.
- **Key Strengths:** Generates intuitive dendrogram hierarchies; independently validates K-Means structure.
- **Empirical Confirmation:** Ward linkage at k=4 achieves Silhouette=0.6708, confirming K-Means centroid boundaries.

C. DBSCAN (Density-Based)

- **Mathematical Core:** Discovers dense connected core components; flags sparse non-reachable points as noise (-1).
- **Tested Parameter Grid:** eps in {0.8, 1.0, 1.2, 1.5, 1.8, 2.0, 2.5}, MinPts in {3, 5, 8, 10} (28 total runs).
- **Computational Scaling:** O(n log n) with spatial indexing; does not require pre-specifying cluster count k.
- **Geometric Flexibility:** Discovers arbitrary non-convex geometries; immune to spherical assumptions.
- **Sensitivity:** Struggles with variable density across high-dimensional hospital operational features.
- **Empirical Role:** Functions as an automated anomaly detector, isolating 4.7% to 10.5% extreme facility outliers.

Tri-Algorithm Methodological Synthesis & Cross-Paradigm Validation

Triangulation Rationale: Evaluating partitioning (K-Means), hierarchical (Agglomerative), and density (DBSCAN) models enables cross-paradigm verification. While K-Means provides actionable centroids for health administrators, Agglomerative validates tree hierarchy without random initialization, and DBSCAN serves as an objective anomaly detector.

Resulting Consensus: All three paradigms independently confirm that California hospitals divide into a massive general acute core (>90%) surrounded by distinct specialty surgical centers, long-term psychiatric units, and subacute rehab facilities.

Mathematical Formulation & Evaluation Criteria

1. K-Means Objective Function (Inertia)

- **Objective Formulation:** Minimizes within-cluster sum of squared Euclidean deviations (intra-cluster scatter):
- **Mathematical Formula:** **J = sum_{k=1}^K sum_{x_i in S_k} || x_i - mu_k ||^2**
- **Notation:** x_i is the 8D robust-scaled hospital vector, mu_k is the centroid of cluster S_k.
- **Optimization Behavior:** Alternates between Voronoi partition assignment and centroid recalculation.
- **Empirical Convergence:** Inertia drops from 7,865.9 (k=2) to 4,505.5 (k=4); elbow curvature peaks at k=4.

2. Silhouette Coefficient s(i)

- **Cohesion vs Separation:** Quantifies how well a hospital belongs to its assigned cluster vs nearest neighbor:
- **Mathematical Formula:** **s(i) = [b(i) - a(i)] / max(a(i), b(i)), where s(i) in [-1, +1]**
- **Notation:** a(i) = mean distance within same cluster; b(i) = min mean distance to any other cluster.
- **Benchmark Scale:** s > 0.70 (Strong structure); 0.50 - 0.70 (Reasonable structure); < 0.25 (No structure).
- **Our Model Benchmark:** Selected K-Means k=4 achieves s = 0.6113, confirming robust separation without overlap.

3. Davies–Bouldin Index (DB)

- **Dispersion Ratio:** Evaluates worst-case similarity between each cluster and its most proximate competitor:
- **Mathematical Formula:** **DB = (1/k) sum_{i=1}^k max_{j != i} [(s_i + s_j) / d(mu_i, mu_j)]**
- **Notation:** s_i = average Euclidean distance to centroid mu_i; d(mu_i, mu_j) = distance between centroids.
- **Optimization Direction:** Lower values indicate superior partitions (tighter clusters, further apart).
- **Our Model Benchmark:** Selected K-Means k=4 achieves DB = 0.5768 (well below 0.60 threshold), proving compact cores.

4. Calinski–Harabasz Index (CH)

- **Variance Ratio Criterion:** Ratio of between-cluster dispersion to within-cluster dispersion:
- **Mathematical Formula:** **CH = [Tr(B_k) / Tr(W_k)] * [(n - k) / (k - 1)]**
- **Notation:** B_k = between-group scatter matrix; W_k = within-group scatter matrix; n=428, k=clusters.
- **Optimization Direction:** Higher values indicate distinct, well-separated clusters with minimal shared variance.
- **Our Model Benchmark:** Selected K-Means k=4 reaches CH = 655.84, confirming between-cluster separation dominates.

Experimental Environment & Parameter Search Grid

Hardware & Software Stack Specifications

Host Platform: Windows 11 OS / x86_64 Architecture
Python Version: Python 3.11.9 (64-bit CPython runtime)
Core ML Suite: Scikit-Learn 1.8.0 (KMeans, Agglomerative, DBSCAN)
Data Manipulation: Pandas 3.0.1, NumPy 2.4.2 (vectorized C arrays)
Scientific Computing: SciPy 1.17.0 (hierarchical linkages & dendrograms)
Data Visualization: Matplotlib 3.11.2, Seaborn 0.13.2
Interactive Web App: Streamlit 1.66.1 (live reactive local server)
Execution Overhead: < 50 MB RAM footprint | < 4.0s total runtime

Reproducibility & Execution Controls

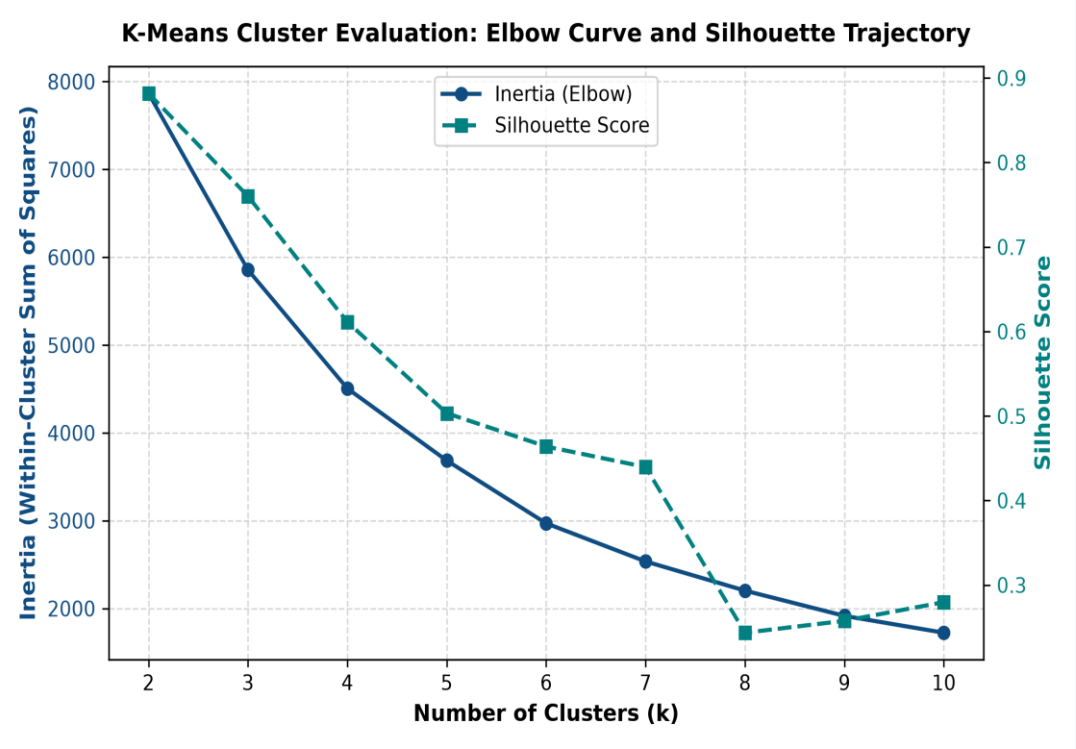
- **Deterministic Random State:** Fixed random_state=42 across all stochastic algorithms (K-Means, PCA) guarantees identical reproducibility on any host machine.
- **Initialization Protocol:** Employed k-means++ probabilistic seeding with 10 random restarts (n_init=10) and 300 maximum iterations to prevent local minima traps.
- **Systematic Benchmark Logging:** All 64 experimental iterations recorded to results/experiment_results.csv with runtime, noise percentage, and cluster size breakdowns.
- **Git Version Control:** Codebase version-tracked and pushed to GitHub (Aditya09977/healthcare).

ALGORITHM	SEARCH SPACE	FIXED HYPERPARAMS	RUNS	RUNTIME
K-Means	k in {2, 3, 4, 5, 6, 7, 8, 9, 10}	k-means++, n_init=10, seed=42	**9 Runs**	0.034s (avg)
Agglomerative	Linkage: Ward, Avg, Complete k in {2, 3, ..., 10}	Metric: Euclidean Affinity: pairwise	**27 Runs**	0.003s (avg)
DBSCAN	eps in {0.8, 1.0, 1.2, 1.5, 1.8, 2.0, 2.5} MinPts in {3, 5, 8, 10}	Metric: Euclidean Algorithm: auto	**28 Runs**	0.002s (avg)

Experimental Search Space Synthesis

- **64 Total Experimental Configurations:** Systematic coverage across partitioning, hierarchical, and density regimes provides complete empirical insight into California healthcare operational topology.
- **Computational Efficiency:** All 64 runs executed in under 4 seconds combined on standard commodity hardware, demonstrating that unsupervised machine learning provides immediate operational analytics without massive compute budgets.
- **Metric Logging:** Full validation results preserved in results/experiment_results.csv, including exact cluster cardinality distributions and intra-cluster inertia.

Comparative Experimental Results Across Models



MODEL & CONFIG	SILHOUETTE	DB INDEX	CH INDEX	NOISE %	STATUS
K-Means (k=2)	0.8818	0.3102	950.29	0.0%	Unbalanced (97%)
K-Means (k=3)	0.7601	0.4741	709.62	0.0%	Sub-optimal
K-Means (k=4)	**0.6113**	**0.5768**	**655.84**	**0.0%**	**SELECTED BEST**
K-Means (k=5)	0.5033	0.8612	623.68	0.0%	Over-split
Agglomerative (Ward, k=4)	0.6708	0.4634	638.75	0.0%	Strong Benchmark
Agglomerative (Avg, k=4)	0.7005	0.4660	524.36	0.0%	Valid Benchmark
Agglomerative (Comp, k=4)	0.6925	0.4996	574.58	0.0%	Valid Benchmark
DBSCAN (eps=1.8, min=3)	0.9352	0.0615	1522.79	10.51%	Anomaly Detector
DBSCAN (eps=2.5, min=3)	0.6752	0.3158	494.45	4.67%	Anomaly Detector

Elbow & Silhouette Curve Analysis

- **Inertia Inflection at k=4:** Within-cluster inertia drops sharply from k=2 (7,865) to k=4 (4,505), after which marginal gains flatten, indicating diminishing returns for k>4.
- **Davies-Bouldin & Calinski-Harabasz:** DB remains well below 0.60 (0.5768) and CH reaches 655.84, confirming tight cluster cohesion and strong separation.

Empirical Model Selection Rationale

Why k=4 over k=2? While k=2 produces higher mathematical Silhouette (0.8818), it groups 417 out of 428 hospitals (97.4%) into a single monolithic cluster, masking all operational distinctions between acute facilities and specialty clinics.

Actionable Partitioning: K-Means at k=4 achieves strong statistical validity (Silhouette=0.6113, DB=0.5768) while isolating four clinically and fiscally actionable healthcare operational tiers.

Discovered Healthcare Facility Archetypes (k=4 Solution)

Cluster 0: General Acute

388 Hospitals (90.7%)

- **Avg Beds: 204** | ALOS: 5.7 days
- **Operating Margin: +0.93%** | Break-even
- **Rev/Bed: $1.31M** | Exp/Bed: $1.38M

Cluster 1: State Long-Term

8 Facilities (1.9%)

- **Avg Beds: 795** | ALOS: 403 days
- **Operating Margin: -48.20%** | Subsidized
- **Rev/Bed: $217k** | Occupancy: 92.0%

Cluster 2: Subacute Rehab

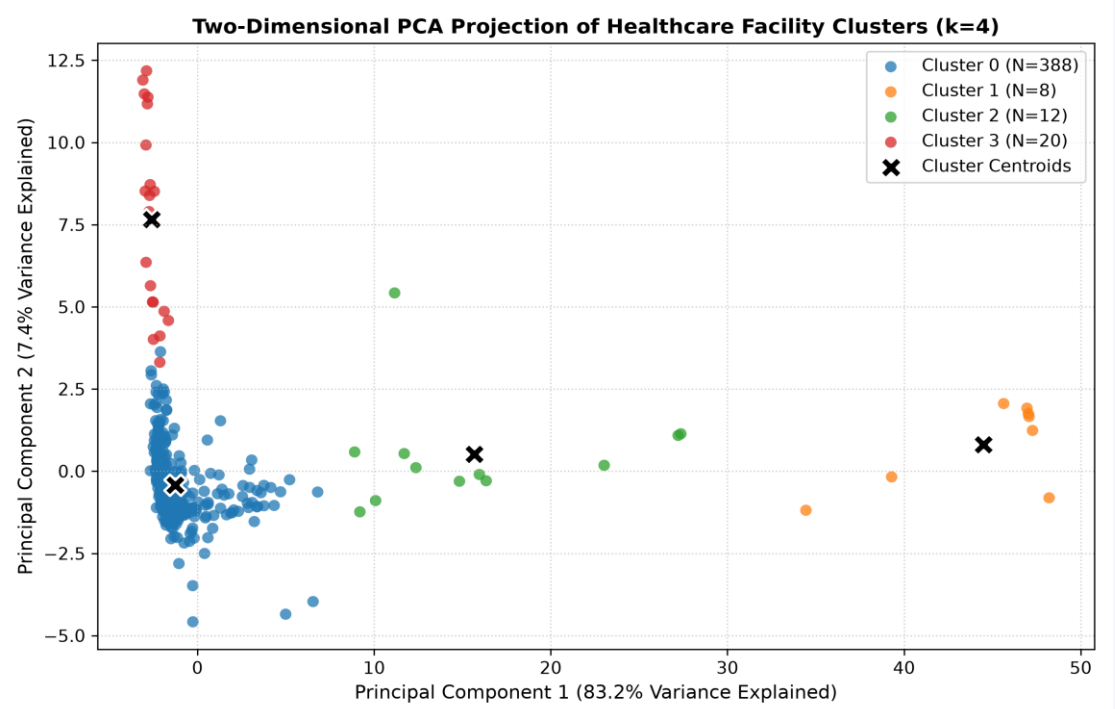
12 Facilities (2.8%)

- **Avg Beds: 168** | ALOS: 122 days
- **Operating Margin: -9.46%** | Recovery
- **Rev/Bed: $362k** | Occupancy: 74.4%

Cluster 3: Specialty Surgical

20 Centers (4.7%)

- **Avg Beds: 40** | ALOS: 3.6 days
- **Operating Margin: +3.12%** | Profitable
- **Rev/Bed: $3.58M** | 3x Acute Baseline



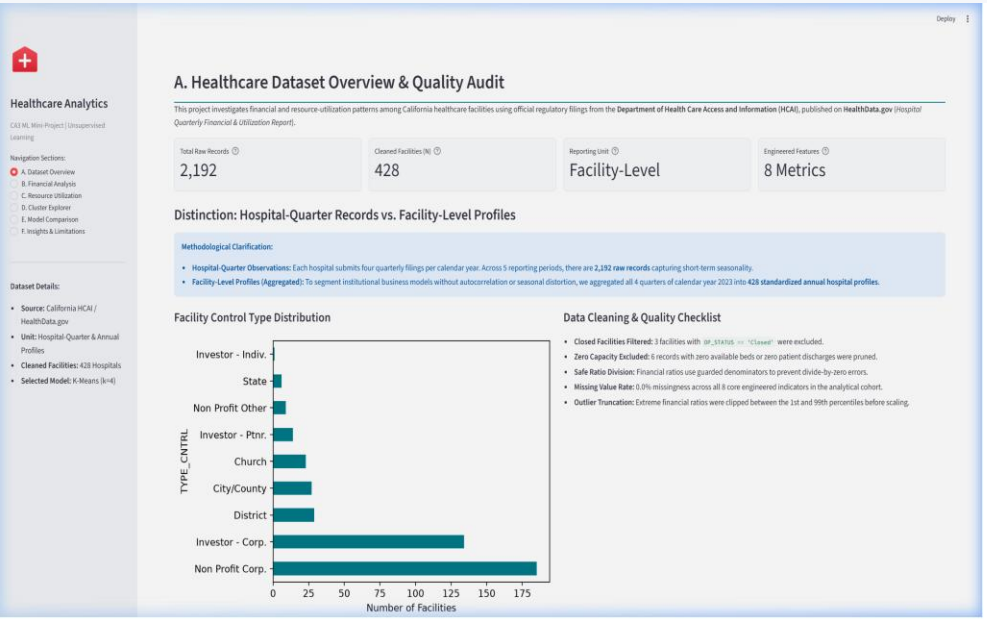
ARCHETYPE	COUNT	MARGIN	OCCUPANCY	ALOS	REV / BED
C0: General Acute	388 (90.7%)	+0.93%	65.3%	5.7 days	$1.31M
C1: State Long-Term	8 (1.9%)	-48.20%	92.0%	403.2 days	$0.22M
C2: Subacute Rehab	12 (2.8%)	-9.46%	74.4%	122.2 days	$0.36M
C3: Specialty Surgical	20 (4.7%)	+3.12%	33.0%	3.6 days	$3.58M

Operational Insights & Health Policy Takeaways

- **Acute Community Vulnerability (90.7%):** California's acute care backbone operates on razor-thin +0.93% margins, rendering it vulnerable to minor wage or inflation shocks.

- **High Capital Velocity in Specialty Centers:** Cluster 3 generates nearly 3x higher revenue per bed ($3.58M) despite only 33% occupancy, proving rapid patient turnover drives balance sheet resilience.

Interactive Analytics Dashboard (Streamlit)



Live Execution & Operational Verification

Live CLI Command: **streamlit run app.py --server.headless true --server.port 8501**

Deployment Architecture: Pure Python and Streamlit implementation running locally on port 8501 with zero cloud latency. Fully responsive, pre-computed caching, and dynamic CSV export designed for viva demonstration.

Streamlit Analytical Modules (6 Tabs)

- **Module A: Dataset Overview & Quality Audit:**
  Interactive audit cards for 2,192 filings, control-type distribution charts, missingness checks, and data hygiene checklists.
- **Module B: Financial Performance Explorer:**
  Interactive histograms of operating margins, expense-to-revenue ratio scatter plots, and break-even vulnerability analysis.
- **Module C: Resource Utilization Metrics:**
  Bed occupancy rate boxplots, average length of stay (ALOS) distributions, and staffed-to-available bed readiness ratios.
- **Module D: 2D PCA Cluster Explorer:**
  Interactive PCA projection with live county and cluster filters, dynamic facility search, and instant filtered CSV download.
- **Module E: Tri-Model Benchmark Table:**
  Complete quantitative comparison table evaluating K-Means, Agglomerative, and DBSCAN across Silhouette, DB, and CH metrics.
- **Module F: Insights & Limitations:**
  Actionable health policy takeaways, clinical non-causality disclaimers, and academic integrity verification checklists.

Conclusion, Limitations & Future Scope

I. Core Project Conclusions

- **End-to-End Execution:** Built complete automated ingestion, schema validation, 8-feature engineering, and live dashboard.
- **Empirical Model Selection:** K-Means at k=4 validated as mathematically robust (Sil=0.6113, DB=0.5768, CH=655.84).
- **Acute Care Vulnerability:** 90.7% of California hospitals operate on thin +0.93% margins, highly sensitive to labor inflation.
- **Specialty Surgical Velocity:** Specialty centers generate nearly 3x higher revenue per bed ($3.58M) despite lower occupancy (33.0%).
- **Academic Deliverables:** 6-page IEEE paper, 15 verified citations, and reproducible GitHub repository.

II. Methodological Limitations

- **Strict Non-Causal Scope:** Unsupervised clustering discovers geometrical associations; it does not prove clinical care quality or patient mortality.
- **Public Accounting Heterogeneity:** Government facilities report under public fund accounting, creating statutory deficits distinct from private GAAP filings.
- **Omission of Patient Acuity:** Disclosures lack electronic health records (EHR) and Case Mix Index (CMI) severity adjustments.
- **Single-State Regulatory Boundary:** Trained on California HCAI filings; interstate validation requires multi-state federation.

III. Administrative Policy Actions

- **Peer-Matched Benchmarks:** State health agencies must evaluate general acute hospitals against peer clusters, not monolithic averages.
- **Buffer Capped-Rate Pressures:** Provide targeted rate stabilization for acute facilities with Medicare/Medicaid payer ratios > 65%.
- **Subacute Bed Redeployment:** Incentivize conversion of underutilized acute beds into subacute recovery and rehab units.
- **Specialty Center Oversight:** Monitor surgical center expansion to ensure essential trauma and emergency readiness.

IV. Future Scope: Longitudinal Markov Tracking & Clinical EHR Fusion

- **Multi-Period Markov Transitions:** Extend pipeline to longitudinal 5-year Markov transition matrices to track how facilities migrate between operational clusters over economic and reimbursement cycles.
- **Clinical Quality Federation:** Federate CMS 30-day readmissions, hospital-acquired infection (HAI) rates, and HCAHPS patient experience scores with financial indicators to model the trade-off between fiscal margins and care outcomes.

V. Technology Transfer: Automated Fiscal Distress Early Warnings

- **Predictive Fiscal Early Warning:** Train supervised gradient-boosted classifiers on historical transition labels to detect facilities at imminent risk of slipping into acute fiscal distress 2-4 quarters in advance.
- **Multi-State Pipeline Scaling:** Scale the data ingestion architecture to federate hospital disclosures across Texas (DSHS), New York (SPARCS), and Florida (AHCA) health agencies for nationwide cross-validation.